

Midterm Exam 1 · Fall 2022

MATH 411 · October 7, 2022

Retypeset past exam for practice. The original allowed one handwritten page of notes, lasted 50 minutes, and was worth 25 points. The original date and Question 3(a) contained errors corrected here.

1. (a) **(3 points)** Find every solution of $z^3 = -8i$. You may leave your answers in polar form $re^{i\theta}$.
(b) **(4 points)** Let $r > 0$ be real and $w \in \mathbb{C}$. Show that

$$\left| \frac{w - r}{1 - r\overline{w}} \right| \leq 1 \quad \text{when } r < 1 \text{ and } |w| < 1.$$

2. **(9 points)** Let $D = \{z \in \mathbb{C} : |z| < 1\}$ and let $f = u + iv$ be holomorphic on D . Suppose $|f|^2 = u^2 + v^2 = c$ throughout D , where $c > 0$. Prove that f is constant.

Hint. Differentiate $u^2 + v^2 = c$ with respect to x and y . You may use that a real function with continuous partial derivatives and both partial derivatives zero on D is constant.

3. **(9 points)** Let f be entire and analytic, and set

$$A_n = \{z \in \mathbb{C} : f^{(n)}(z) = 0\}, \quad n \geq 0.$$

- (a) Show that A_n is closed. Show also that A_n is discrete if $f^{(n)}$ is not identically zero. What is A_n if $f^{(n)}$ is identically zero?

Here *discrete* means that every point of the set has a neighborhood containing no other point of the set.

- (b) Suppose that for every $a \in \mathbb{C}$, at least one coefficient in the Taylor expansion

$$f(z) = \sum_{n \geq 0} c_n(a)(z - a)^n$$

is zero. Show that f is a polynomial. You may use part (a) and the fact that $\mathbb{C}$ is uncountable.